

Board of Studies (BoS) Syllabus

Course Information

Degree: UG

Program: BCA

Course Name: Computer Organization

Course Credits: 04

Semester: 1st Sem

Course Category: Core

Type of Course: Lecture / Practical / Experiential

New Course: yes

Total Hours:30Hours

Learning Hours

Activity	Hours per Week
Lecture	02
Tutorial	0
Practical	02
Experiential Learning	03

Course Contribution

Category	Percentage Contribution
Employment Enhancement	30
Entrepreneurship Skills	30
Skill Development	40

Course Relevance

Level	Percentage Relevance
Global	20
National	30
Regional	40
Local	10

Course Outcomes (COs)

CO#	Course Outcome	Bloom's Taxonomy Level
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CO1	To Understand the basics of Digital Electronics and Binary Number System	BL2			
CO2	To Learn the implementation of Combinational Circuit.	BL6			
CO3	To Learn the implementation of Sequential Circuit.	BL6			
CO4	To Understand the Organization of basic computers.	BL2			
CO5	To Understand the concept of memory management	BL2			
Module	Course Content	Tools Used / Assessment and Activity	Bloom's Level	Mapping COs	Percentage change in Module (put Nil if there is no change or it is a new course write NEW)
Module 1: Digital Principles: 6 Hours	Definition for Digital signals, Digital logic, Digital computers	presentation	BL-2	1	NEW
	Von Neumann Architecture, Boolean Laws and Theorems	presentation	BL-4	1	NEW
	K-Map: Truth Tables to K-Map, 2, 3 and 4 variable K Map	Quiz	BL-3	1	NEW
	K-Map Simplifications, Don't Care Conditions, SOP and POS.	Simulators	BL-2	1	NEW
	Number Systems: Decimal, Binary, Octal, Hexadecimal, Number System Conversions,	presentation	BL-3	1	NEW
	Binary Arithmetic, Addition and subtraction of BCD	presentation	BL-4	1	NEW
	Octal Arithmetic, Hexadecimal Arithmetic, Binary Codes,	Simulation	BL-3	1	NEW
	Decimal Codes, Error detecting and correcting codes,	Problem solving	BL-3	1	NEW
Module 2: Combinational Circuits: 6 Hours	Half Adder and Full Adder	presentation	BL-2	2	NEW
	Subtractor, Decoders, Encoder,	presentation	BL-4	2	NEW
	Multiplexer, Demultiplexer	Problem solving	BL-3	2	NEW

	Sequential Circuits: Flip-Flops- SR Flip- Flop	Problem solving	BL-2	2	NEW
	D Flip-Flop, J-K Flip-Flop, T Flip-Flop	Simulation	BL-3	2	NEW
	Register: 4 bit register with parallel load, Shift register-Bidirectional shift register with parallel load	Emulators	BL-4	2	NEW
Module 3:Basic Computer Organization and Design: 6 Hours	Instruction Codes, Computer Registers, Computer Instruction	presentation	BL-2	3	NEW
	Timing and Control, Instruction Cycle	Hands on lab	BL-4	3	NEW
	Memory-Reference Instructions, Input-Output Interrupt	Hands on lab	BL-3	3	NEW
	Complete Computer Description, Design of Basic Computer,	Hands on lab	BL-2	3	NEW
	Design of Accumulator logic. Central Processing Unit: Introduction, General Register Organization, Stack	Simulator	BL-3	3	NEW
	Organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation,	Simulator	BL-4	3	NEW
Module 4: Pipeline and Vector Processing: 6 Hours	Parallel Processing, Pipelining	Case studies	BL-2	4,5	NEW
	Arithmetic Pipeline	Case studies	BL-4	4,5	NEW
	RISC Pipeline. Input-Output Organization:	Case studies	BL-3	4,5	NEW
	Peripheral Devices, Input-Output Interface,	Case studies	BL-2	4,5	NEW
	Asynchronous data transfer	presentation	BL-3	4,5	NEW
	Modes of Transfer, Priority Interrupt	presentation	BL-4	4,5	NEW
Module 5:	Memory Hierarchy, Main Memory	experiments	BL-2	6	NEW
	Auxiliary memory	Group discussion	BL-3	6	NEW
	Associate Memory	experiments	BL-2	6	NEW

Memory Organization:	Cache Memory	Case study	BL-4	6	NEW
6 Hours	Virtual Memory	Case study	BL-2	6	NEW
	Memory Management Hardware	Case study	BL-2	6	NEW

CO-PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
C01	2	-	-	2	-	-	-	-	-
C02	-	3	3	2	3	-	2	-	-
C03	-	-	3	-	3	-	-	3	-
C04	-	2	-	-	-	-	2	-	-
C05	-	2	-	3	-	-	2	-	-
C06	-	-	-	-	2	-	2	-	-

Text Books:

1. Donald P Leach, Albert Paul Malvino, Goutam Saha- “Digital Principles & Applications” , Tata McGraw Hill Education Private Limited, 2011 Edition.
2. M. Morris Mano- “Computer System Architecture”, Pearson/Phi, Third Edition.

Reference Books:

- 1 William Stallings- “Computer Organization and Architecture”, Pearson/PHI, Sixth Edition,
- 2 Andrew S. Tanenbaum- “Structured Computer Organization”, PHI /Pearson 4th Edition,
- 3 M.V .Subramanyam, “Switching Theory and Logic Design”, Laxmi Publications (P)Ltd.
- 4 Ikinderpal Singh, Computer Organization Architecture, Khanna Book Publishing.

Suggestive Laboratory Experiments:

1. Verify logic behavior of AND, OR, NAND, NOR, EX-OR, EX-NOR, Invert and Buffer gates.
2. To study and verify NAND as a Universal Gate

3. To verify De- Morgan's theorem for 2 variables
4. Design and test of an S-R flip-flop using NAND/NOR gate.
5. Convert BCD to Excess-3 code using NAND gate
6. To Convert Binary to Grey Code
7. Verification of Truth Tables of J-K Flip-Flop using NAND/NOR gate
8. Realize Decoder and Encoder circuit using Basic Gates.
9. Design and implement the 4:1 MUX using gates.
10. Implementation of 4-Bit Parallel Adder Using 7483 IC.
11. Design and verify operation of half adder and full adder.
12. Design and verify operation of half subtractor.
13. Design and implement a 4 bit shift register using Flip flops.
14. Implement Boolean function using logic gates in both SOP and POS
15. Design and implement a 4 bit synchronous counter.
16. Design and verify 4 bit asynchronous counter.

Hardware

1. Familiarize the computer system layout: marking positions of SMPS, motherboard, FDD, HDD, CD, DVD and add on cards.
2. Identify the Computer Name and Hardware Specification (RAM capacity, Processor type, HDD, 32 bit/ 64 bit)
3. Identify and Trouble shoot the problems of RAM, SMPS and motherboard
4. Configure BIOS settings- disable and enable USB and LAN
5. Adding additional RAM to the system.(expanding RAM size).
6. To Study mother board layout of a system.
7. Demonstrate the assembly of a PC
8. Demonstration of various ports: CPU, VGA port, PS/2 (keyboard, mouse) USB, LAN, Speaker, Audio.
9. Install and configure windows OS

10. To study the installation of Printer and trouble shooting.